REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

Four-slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM. FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

U1
Pico 2 with headers, SC1632

| | | | |
|---|---|---|---|
| P_S1_IRQ | 1 | GP0 | GP28 | 21 | P_EXP_RESET_N |
| P_S1_BUSY | 2 | GP1 | GP27 | 22 | P_IO_OE_N |
| GND | 3 | GND | GND | 23 | GND |
| P_S2_IRQ | 4 | GP2 | GP26 | 24 | |
| P_S2_BUSY | 5 | GP3 | GP19 | 25 | |
| P_SPI_MISO | 6 | GP4 | GP20 | 26 | |
| P_S1_CS_N | 7 | GP5 | GP21 | 27 | |
| GND | 8 | GND | GND | 28 | GND |
| P_SPI_SCK | 9 | GP6 | GP22 | 29 | |
| P_SPI_MOSI | 10 | GP7 | RUN | 30 | |
| P_S3_IRQ | 11 | GP8 | GP25 | 31 | |
| P_S3_BUSY | 12 | GP9 | GP27 | 32 | |
| GND | 13 | GND | AGND | 33 | GND |
| P_S4_IRQ | 14 | GP10 | GP28 | 34 | |
| P_S4_BUSY | 15 | GP11 | ADC_VREF | 35 | |
| P_S2_CS_N | 16 | GP12 | 3V3_OUT | 36 | PICO_3V3 |
| P_S3_CS_N | 17 | GP13 | 3V3_EN | 37 | |
| GND | 18 | GND | GND | 38 | GND |
| P_S4_CS_N | 19 | GP14 | VSYS | 39 | PICO_VSYS |
| P_EXP_CS_N | 20 | GP15 | VBUS | 40 | USB_VBUS |

J1
SSQ-110-01-G-S

| | | | |
|---|---|---|---|
| GND | 1 | 1 | 6 | SPI_MOSI |
| RADIO_5V | 2 | 2 | 7 | SPI_MISO |
| RADIO_5V | 3 | 3 | 8 | S1_CS_N |
| GND | 4 | 4 | 9 | S1_RESET_N |
| SPI_SCK | 5 | 5 | 10 | GND |

J3
SSQ-110-01-G-S

| | | | |
|---|---|---|---|
| GND | 1 | 1 | 6 | SPI_MOSI |
| RADIO_5V | 2 | 2 | 7 | SPI_MISO |
| RADIO_5V | 3 | 3 | 8 | S2_CS_N |
| GND | 4 | 4 | 9 | S2_RESET_N |
| SPI_SCK | 5 | 5 | 10 | GND |

J9
5 V / 4 A INPUT

| | | | |
|---|---|---|---|
| EXT_5V | 1 | 1 | 2 | GND |

U2
TPS25947OLRPWR

| | | | |
|---|---|---|---|
| UVLO | 1 | EN_UVLO | OUT | 6 | RADIO_5V |
| OVLO | 2 | OVLO | DVDT | 7 | DVDT |
| | 3 | AUXOFF | GND | 8 | GND |
| | 4 | FLT_N | ILIM | 9 | ILIM |
| EXT_5V | 5 | IN | ITIMER | 10 | |

Q1
DMG2305UX-7

| | | | |
|---|---|---|---|
| USB_VBUS | 1 | G | D | 3 | RADIO_5V |
| PICO_VSYS | 2 | S | | | |

J2
SSQ-108-01-G-S

| | | | |
|---|---|---|---|
| S1_BUSY | 1 | 1 | 5 | |
| S1_IRQ | 2 | 2 | 6 | |
| S1_TX_EN | 3 | 3 | 7 | IOREF_3V3 |
| S1_RX_EN | 4 | 4 | 8 | GND |

J4
SSQ-108-01-G-S

| | | | |
|---|---|---|---|
| S2_BUSY | 1 | 1 | 5 | |
| S2_IRQ | 2 | 2 | 6 | |
| | 3 | 3 | 7 | IOREF_3V3 |
| | 4 | 4 | 8 | GND |

U3
AP2112K-3.3TRG1

| | | | |
|---|---|---|---|
| RADIO_5V | 1 | VIN | NC | 4 | |
| GND | 2 | GND | VOUT | 5 | IOREF_3V3 |
| RADIO_5V | 3 | EN | | | |

J10
SSQ-120-01-G-S

| | | | |
|---|---|---|---|
| P_S1_IRQ | 1 | 1 | 11 | P_S3_IRQ |
| P_S1_BUSY | 2 | 2 | 12 | P_S3_BUSY |
| GND | 3 | 3 | 13 | GND |
| P_S2_IRQ | 4 | 4 | 14 | P_S4_IRQ |
| P_S2_BUSY | 5 | 5 | 15 | P_S4_BUSY |
| P_SPI_MISO | 6 | 6 | 16 | P_S2_CS_N |
| P_S1_CS_N | 7 | 7 | 17 | P_S3_CS_N |
| GND | 8 | 8 | 18 | GND |
| P_SPI_SCK | 9 | 9 | 19 | P_S4_CS_N |
| P_SPI_MOSI | 10 | 10 | 20 | P_EXP_CS_N |

J5
SSQ-110-01-G-S

| | | | |
|---|---|---|---|
| GND | 1 | 1 | 6 | SPI_MOSI |
| RADIO_5V | 2 | 2 | 7 | SPI_MISO |
| RADIO_5V | 3 | 3 | 8 | S3_CS_N |
| GND | 4 | 4 | 9 | S3_RESET_N |
| SPI_SCK | 5 | 5 | 10 | GND |

J7
SSQ-110-01-G-S

| | | | |
|---|---|---|---|
| GND | 1 | 1 | 6 | SPI_MOSI |
| RADIO_5V | 2 | 2 | 7 | SPI_MISO |
| RADIO_5V | 3 | 3 | 8 | S4_CS_N |
| GND | 4 | 4 | 9 | S4_RESET_N |
| SPI_SCK | 5 | 5 | 10 | GND |

U4
MCP23S17-E/SO

| | | | |
|---|---|---|---|
| S1_RESET_N | 1 | GPB0 | A2 | 15 | GND |
| S2_RESET_N | 2 | GPB1 | A1 | 16 | GND |
| S3_RESET_N | 3 | GPB2 | A0 | 17 | GND |
| S4_RESET_N | 4 | GPB3 | RESET_N | 18 | EXP_RESET_N |
| | 5 | GPB4 | INTB | 19 | |
| | 6 | GPB5 | INTA | 20 | |
| | 7 | GPB6 | GPA0 | 21 | S1_TX_EN |
| | 8 | GPB7 | GPA1 | 22 | S1_RX_EN |
| IOREF_3V3 | 9 | VDD | GPA2 | 23 | S3_TX_EN |
| GND | 10 | VSS | GPA3 | 24 | S3_RX_EN |
| EXP_CS_N | 11 | CS_N | GPA4 | 25 | S4_TX_EN |
| SPI_SCK | 12 | SCK | GPA5 | 26 | S4_RX_EN |
| SPI_MOSI | 13 | SI | GPA6 | 27 | |
| SPI_MISO | 14 | SO | GPA7 | 28 | |

J6
SSQ-108-01-G-S

| | | | |
|---|---|---|---|
| S3_BUSY | 1 | 1 | 5 | |
| S3_IRQ | 2 | 2 | 6 | |
| S3_TX_EN | 3 | 3 | 7 | IOREF_3V3 |
| S3_RX_EN | 4 | 4 | 8 | GND |

J8
SSQ-108-01-G-S

| | | | |
|---|---|---|---|
| S4_BUSY | 1 | 1 | 5 | |
| S4_IRQ | 2 | 2 | 6 | |
| S4_TX_EN | 3 | 3 | 7 | IOREF_3V3 |
| S4_RX_EN | 4 | 4 | 8 | GND |

#FLG01
POWER DECLARATION: EXT_5V

| | | |
|---|---|---|
| EXT_5V | 1 | POWER |

#FLG02
POWER DECLARATION: GND

| | | |
|---|---|---|
| GND | 1 | POWER |

#FLG03
POWER DECLARATION: PICO_VSYS

| | | |
|---|---|---|
| PICO_VSYS | 1 | POWER |

J11
SSQ-120-01-G-S

| | | | |
|---|---|---|---|
| P_EXP_RESET_N | 1 | 1 | 11 | |
| P_IO_OE_N | 2 | 2 | 12 | |
| GND | 3 | 3 | 13 | GND |
| | 4 | 4 | 14 | |
| | 5 | 5 | 15 | |
| | 6 | 6 | 16 | PICO_3V3 |
| | 7 | 7 | 17 | |
| GND | 8 | 8 | 18 | GND |
| | 9 | 9 | 19 | PICO_VSYS |
| | 10 | 10 | 20 | USB_VBUS |

U5
SN74LVC8T245PWR

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | VCCA | GND | 13 | GND |
| PICO_3V3 | 2 | DIR | B8 | 14 | EXP_RESET_N |
| P_SPI_SCK | 3 | A1 | B7 | 15 | EXP_CS_N |
| P_SPI_MOSI | 4 | A2 | B6 | 16 | S4_CS_N |
| P_S1_CS_N | 5 | A3 | B5 | 17 | S3_CS_N |
| P_S2_CS_N | 6 | A4 | B4 | 18 | S2_CS_N |
| P_S3_CS_N | 7 | A5 | B3 | 19 | S1_CS_N |
| P_S4_CS_N | 8 | A6 | B2 | 20 | SPI_MOSI |
| P_EXP_CS_N | 9 | A7 | B1 | 21 | SPI_SCK |
| P_EXP_RESET_N | 10 | A8 | OE_N | 22 | P_IO_OE_N |
| GND | 11 | GND | VCCB | 23 | IOREF_3V3 |
| GND | 12 | GND | VCCB | 24 | IOREF_3V3 |

U6
SN74LVC8T245PWR

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | VCCA | GND | 13 | GND |
| GND | 2 | DIR | B8 | 14 | S4_IRQ |
| P_S1_BUSY | 3 | A1 | B7 | 15 | S4_BUSY |
| P_S1_IRQ | 4 | A2 | B6 | 16 | S3_IRQ |
| P_S2_BUSY | 5 | A3 | B5 | 17 | S3_BUSY |
| P_S2_IRQ | 6 | A4 | B4 | 18 | S2_IRQ |
| P_S3_BUSY | 7 | A5 | B3 | 19 | S2_BUSY |
| P_S3_IRQ | 8 | A6 | B2 | 20 | S1_IRQ |
| P_S4_BUSY | 9 | A7 | B1 | 21 | S1_BUSY |
| P_S4_IRQ | 10 | A8 | OE_N | 22 | P_IO_OE_N |
| GND | 11 | GND | VCCB | 23 | IOREF_3V3 |
| GND | 12 | GND | VCCB | 24 | IOREF_3V3 |

U7
SN74LVC1T45DBVR

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | VCCA | B | 6 | SPI_MISO |
| GND | 2 | GND | DIR | 5 | GND |
| P_SPI_MISO | 3 | A | VCCB | 4 | IOREF_3V3 |

MP1
M3 captive-post clamp nut

MP2
M3 captive-post clamp nut

MP3
M3 captive-post clamp nut

MP4
M3 captive-post clamp nut

MP5
External cable mating plug

R1
267k

| | | | |
|---|---|---|---|
| EXT_5V | 1 | 1 | 2 | UVLO |

R2
100k

| | | | |
|---|---|---|---|
| UVLO | 1 | 1 | 2 | GND |

R3
340k 0.1%

| | | | |
|---|---|---|---|
| EXT_5V | 1 | 1 | 2 | OVLO |

R4
100k 0.1%

| | | | |
|---|---|---|---|
| OVLO | 1 | 1 | 2 | GND |

R5
715

| | | | |
|---|---|---|---|
| ILIM | 1 | 1 | 2 | GND |

C1
10nF

| | | | |
|---|---|---|---|
| DVDT_C | 1 | 1 | 2 | GND |

C2
1uF

| | | | |
|---|---|---|---|
| EXT_5V | 1 | 1 | 2 | GND |

C3
220uF

| | | | |
|---|---|---|---|
| RADIO_5V | 1 | + | 2 | GND |

C4
22uF

| | | | |
|---|---|---|---|
| RADIO_5V | 1 | 1 | 2 | GND |

C5
22uF

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

C6
100nF

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

C7
100nF

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | GND |

C8
100nF

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

C9
100nF

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | GND |

C10
100nF

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

C11
100nF

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | GND |

C12
100nF

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

R6
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_IO_OE_N |

R7
100k

| | | | |
|---|---|---|---|
| USB_VBUS | 1 | 1 | 2 | GND |

R8
10k

| | | | |
|---|---|---|---|
| EXP_RESET_N | 1 | 1 | 2 | GND |

R9
10k

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | EXP_CS_N |

R10
100k

| | | | |
|---|---|---|---|
| SPI_MISO | 1 | 1 | 2 | GND |

R11
1k

| | | | |
|---|---|---|---|
| RADIO_5V | 1 | 1 | 2 | GND |

R12
1k

| | | | |
|---|---|---|---|
| IOREF_3V3 | 1 | 1 | 2 | GND |

R13
10k

| | | | |
|---|---|---|---|
| P_EXP_RESET_N | 1 | 1 | 2 | GND |

R14
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_S1_CS_N |

R15
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_S2_CS_N |

R16
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_S3_CS_N |

R17
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_S4_CS_N |

R18
10k

| | | | |
|---|---|---|---|
| PICO_3V3 | 1 | 1 | 2 | P_EXP_CS_N |

R19
100

| | | | |
|---|---|---|---|
| DVDT | 1 | 1 | 2 | DVDT_C |

18/26 GPIO used; eight spare including GP26–28, all physically NC.
BUSY/IRQ/CS are direct GPIO through fixed-direction isolation, not expander GPIO.
KEY: captive M3 x 25 alignment post, see radio-slot-v0.1.md.
U4 GPA0..5 = S1/S3/S4 TX/RX pairs; GPB0..3 = S1..4 RESET_N.
No autonomous duty-cycled RX or automatic TX/RX transitions on Ebyte cards.

USB-only: Q1 off; RADIO_5V and IOREF_3V3 unpowered. External power: U2 -> radios and Q1 -> VSYS.
Q1 orientation is intentional: body diode from RADIO_5V toward VSYS. Pico onboard D1 blocks USB backfeed.
#FLG01/#FLG02 model the external supply; #FLG03 models real Q1/onboard USB-D1 ORing output.
U2 AUXOFF/FLT/ITIMER unused per datasheet. Leave ITIMER open for fastest current limiting.

H1
Captive M3 x 25 alignment post

H2
Captive M3 x 25 alignment post

H3
Captive M3 x 25 alignment post

H4
Captive M3 x 25 alignment post

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: carrier-rev-a.kicad_sch

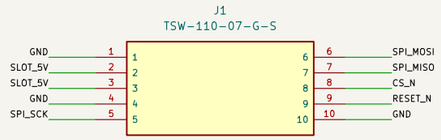
**Title: Four-slot carrier: protected external 5 V / 4 A, isolated GPIO**

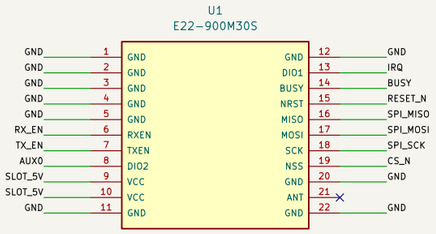
| | | |
|---|---|---|
| Size: A1 | Date: 2026-09-17 | Rev: A-schematic |
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |

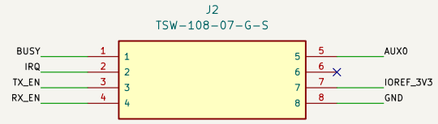
REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

915 MHz LoRa card

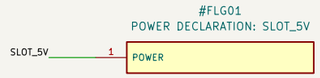
Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

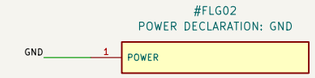


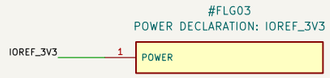



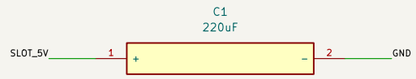



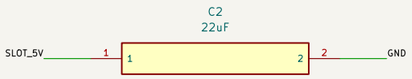

Module pad 21 ANT: NC, use module IPEX. Both pads 9 and 10 are VCC. H1 is the mandatory alignment-post hole.
DIO2 is AUX0 for a local test point only; do not connect it to a Pico GPIO.
AUX1 is reserved. DIO3 is internal TCXO control, configured to 2.2 V.

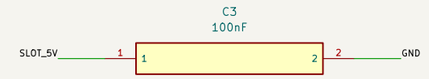


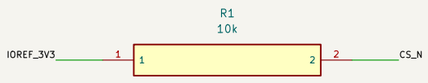



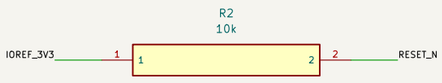



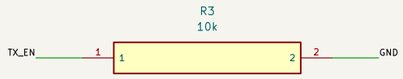



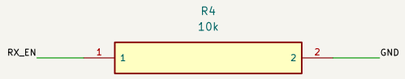



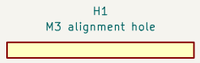




KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
No PCB layout is authorized. Footprint and assembly evidence: see DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e22-900m30s-rev-a.kicad_sch

**Title: 915 MHz LoRa card**

| Size: A3 | Date: 2026-09-17 | **Rev: A-schematic** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |

REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

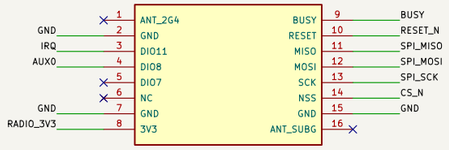
915 MHz FLRC socket adapter

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

J1
TSW-110-07-G-S




U1
Core2021-HF-M plug-in




U2
AP2112K-3.3TRG1




J2
TSW-108-07-G-S




U1 represents the plug-in module, not a soldered module footprint.
Socket contacts for module pins 1, 5, 6 and 16 are isolated.
TX_EN/RX_EN slot contacts unused. No external RF switch added.
J3/J4: 15.24 mm row spacing; top/bottom numbering in FOOTPRINT_REVIEW.md.

#FLG01
POWER DECLARATION: SLOT_5V




#FLG02
POWER DECLARATION: GND




C1
22uF




C2
22uF




C3
100nF




J3
SSQ-108-01-G-S




J4
SSQ-108-01-G-S




C4
22uF




R1
10k




R2
10k




H1
M3 alignment hole



KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
No PCB layout is authorized. Footprint and assembly evidence: see DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: core2021-hf-m-rev-a.kicad_sch

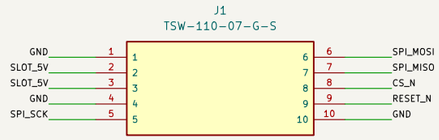
**Title: 915 MHz FLRC socket adapter**

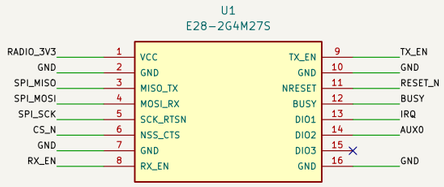
| Size: A3 | Date: 2026-09-17 | **Rev: A-schematic** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |

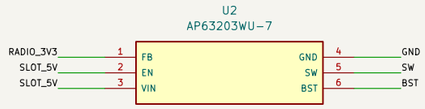
REV A SCHEMATIC REVIEW. NO PCB LAYOUT OR FABRICATION RELEASE.

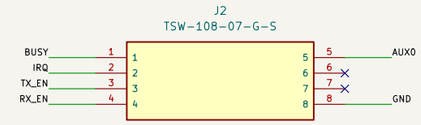
2.4 GHz LoRa or FLRC card, same hardware

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

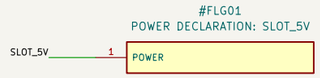


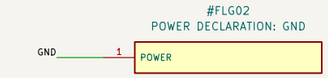



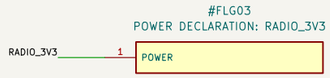



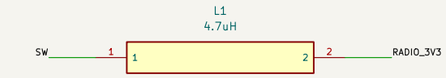



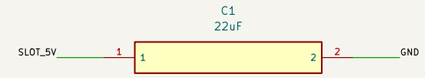

ASSEMBLY VARIANT: IPEX ONLY. Selector MUST match manual section 9.2 right image.
MCP23S17 controls TX_EN/RX_EN before RF commands. No automatic RX/TX modes.
Buck PCB layout must follow figure 25, 2 oz copper on both layers.

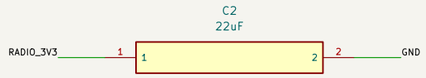


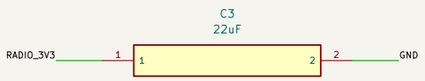



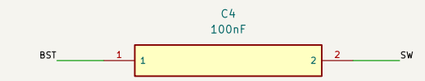



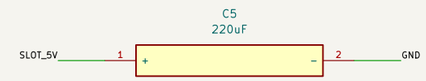



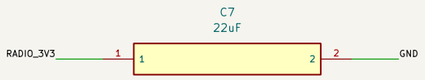



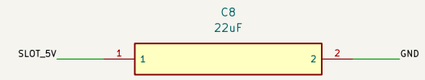



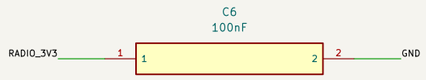



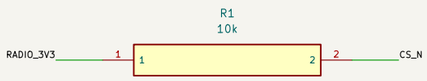



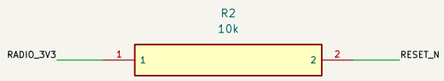



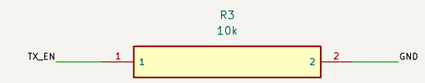










KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
No PCB layout is authorized. Footprint and assembly evidence: see DESIGN_REVIEW.md.

SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: e28-2g4m27s-rev-a.kicad_sch

**Title: 2.4 GHz LoRa or FLRC card, same hardware**

| Size: A3 | Date: 2026-09-17 | **Rev: A-schematic** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |